Demo Study C: A Physics-Informed Hybrid Model for Link Margin Forecasting

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SYNTHETIC DEMO DOCUMENT - fictional content for software testing. Not a real publication.

Abstract

This study introduces a hybrid forecasting approach that combines the ITU-R rain model with a gated recurrent unit (GRU) residual learner. Probabilistic prediction intervals are produced via quantile regression.

Introduction

Pure data-driven models can violate physical constraints. We develop a hybrid model that embeds physics-based attenuation estimates into a neural network, and we address predictive uncertainty explicitly.

Data

We use a multi-year beacon dataset of 4 years from two stations at Ka-band and Ku-band. Seasonal variations were studied through year-round evaluation.

Methodology

The hybrid approach adds a GRU residual term to the ITU-R P.618 estimate and is trained with a quantile loss for 90 percent prediction intervals. Evaluation uses RMSE, outage probability, and calibration of prediction intervals.

Results

The hybrid model achieves an RMSE of 0.71 dB and reduces outage probability error by 24 percent compared with the physics baseline. Prediction interval coverage reaches 89 percent for the nominal 90 percent interval.

Limitations

Real-time on-board deployment was not investigated, and inference latency was not measured. Only two stations were used, which limits conclusions about generalization to other regions.